F16GE104-CB

400A
4 POLE

G
3~

F16-GE-104
415V
250kVA

F16-SR-002
415V FLOTATION SWITCHROOM

SWR EARTH BAR

FEEDER FROM F16-SW-004
2000-F16-SLD-EL-10201
TO F16-SW-004
2000-F16-SLD-EL-10201

F16-SW-104
FLOTATION 415V ESSENTIAL SWITCHBOARD

EARTH BAR

NEUTRAL BAR
NOTE 1

AFD

125A
160A

SD
NOTE 3

2A
63A

2A
63A

PFR

2A
63A

MANUAL
MCCB CHANGEOVER
SWITCH 3 POLE 400A

415V 3NE~ 50Hz 400A 65kA for 1sec

00F-05
INC

01F-04
63A
2A
32A
70A
63A
63A

01F-09
250A
2A
32A
90A

1F-01
250A
2A
32A
90A

01F-13
250A
2A
32A
90A

ISL

30kW

F16UPS002-ISL
415VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F16-SCM-EL-10215

F16-DB-202
FLOTATION
415V SWITCHROOM
415VAC UPS DB
2000-F16-TLD-EL-10056

UPS
30kVA

F16-UPS-002
FLOTATION
415V SWITCHROOM
415VAC UPS
2000-F16-TLD-EL-10056

JB F17DB301-JB

JB F18DB301-JB

JB F16DB301-JB

69kW
F17-DB-301
FLOTATION
TAILINGS THICKENER AREA
L&SP DB
2000-F17-SCM-EL-10201

69kW
F18-DB-301
CLEANER
FLOTATION AREA
L&SP DB
2000-F18-SCM-EL-10201

69kW
F16-DB-301
FLOTATION AREA
L&SP DB
2000-F16-SCM-EL-10205

F16-SR-001
11KV FLOTATION
SWITCHROOM

15kW
F16-DB-201
FLOTATION
11KV SWITCHROOM
415VAC UPS DB
2000-F16-TLD-EL-10057

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM F16-SW-004 INCOMER TIER. SIGNAGE SHALL BE PLACED ON F16-SW-104 SWITCHBOARD AND ON GENERATOR READING "M.E.N. LINK LOCATED IN F16-SW-004 INCOMER TIER"
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
4. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
5. CIRCUIT BREAKER TO BE SUBSTITUTED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.

LEGEND

PFR	PHASE FAILURE RELAY
I >	EARTH LEAKAGE RELAY NOTE 4, 5
xxxA / xxxA	ST TRIP SET POINT / CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY

DRG No.	REFERENCES
2000-F16-SCM-EL-10224	F16-SW-004 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F16-SCM-EL-10223	F16-SW-004 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F16-SCM-EL-10216	F16-SW-104 415V ESSENTIAL SWITCHBOARD INCOMER SCHEMATIC DIAGRAM
2000-F16-DRG-EL-10101	F16-SR-002 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F16-BLK-IC-10001	F16-SR-002 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F16-SLD-EL-10201	F16-SW-004 415V MCC SINGLE LINE DIAGRAM SHEET 1
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	07SEP26	KCGM GROWTH PROJECT - SP1 AS-BUILT					
1	26AUG25	FAT REDLINES INCORPORATED					
0	29AUG24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: ROUGHER AND SCAVENGER FLOTATION
F16-SW-104 415V ESSENTIAL SWITCHBOARD
SINGLE LINE DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F16-SLD-EL-10251	2

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235TI KCGM GROWTH PROJECT\03 DELIVERABLES\ELS\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SLD-EL-10251.DWG 03SEP26 3:07PM ROBINLAW